

FragmentIQ: The Systematic Bias of Image-Based Muckpile Fragmentation Analysis, and Its Correction

Felipe Santibañez-Leal 

Abstract—How well a blast fragmented the rock, the size distribution of the muckpile, sets the energy cost of every crushing and grinding stage downstream, and it is routinely estimated by photographing the muckpile and analysing the image. Such image-based granulometry is convenient and its biases are well known but rarely quantified against ground truth, because on a real muckpile the true size distribution is unknown. FragmentIQ closes that loop on synthetic muckpiles whose true fragmentation is known by construction: it runs the classical watershed delineation and Rosin-Rammler fit of the standard tools, and measures the error. The result is a clean, quantified statement of the two biases. First, the watershed *over-segments*: it finds one-and-a-half to more than two times as many fragments as truly exist, splitting single blocks into pieces, and as a consequence the recovered P_{50} and P_{80} fall systematically below the truth, by roughly twenty to thirty percent across seven cases. Second, and correctable, a size calibration that accounts for the missed fines and the over-segmentation reduces the median P_{50} error from 28.4% to 3.98%, and a learned convolutional edge-delineation, in place of the classical watershed, trims it modestly further (from 27.2% to 23.8% with a boundary F1 of 0.997). The honest lesson is that a raw image-based fragmentation number is biased fine by tens of percent and should be calibrated before it is trusted, and the tool makes that bias and its correction visible and reproducible; every number and figure regenerates from the committed artifacts.

Index Terms—Rock fragmentation, muckpile, particle size distribution, image analysis, watershed segmentation, Rosin-Rammler, Kuz-Ram, blasting, calibration, reproducibility.

I. INTRODUCTION

THE degree to which a drill-and-blast round fragments the rock, quantified as the particle-size distribution of the resulting muckpile, is one of the most consequential numbers in a mine: a finer muckpile is cheaper to dig, haul and, above all, crush and grind, so blast design is tuned to it, and predicting it from blast parameters is the classic Kuz-Ram problem [1], [2]. Measuring it after the fact, by sieving, is impractical at production scale, so the standard practice is image-based granulometry: photograph the muckpile, delineate the visible fragments, convert their areas to sizes, and fit a Rosin-Rammler distribution [3]. The commercial tools that do this (WipFrag, Split) are built on a watershed segmentation of the image [5], [4], and their biases, over-segmentation of large blocks, blindness to fines below the image resolution, are well

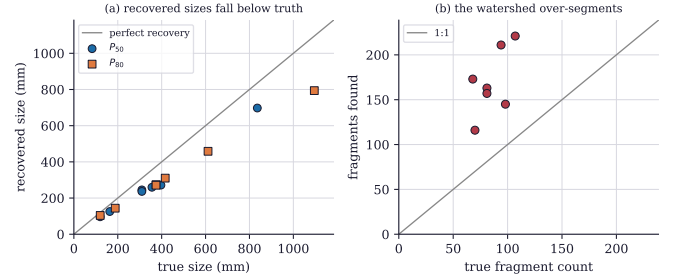


Fig. 1. The bias of image-based fragmentation analysis, measured against known ground truth over seven synthetic muckpiles. (a) Recovered versus true P_{50} and P_{80} : every point falls below the line of perfect recovery, so the watershed pipeline reports sizes systematically finer than truth, by roughly 20–30%, and the gap widens for the coarse tail. (b) The cause: fragments found versus true fragment count, every point above the one-to-one line, the watershed *over-segments*, finding 1.5 to more than 2 times as many fragments as exist by splitting single blocks, which shifts the size distribution fine.

documented in the literature and usually acknowledged with a warning rather than a number, because on a real muckpile the true distribution against which to measure the error is not known.

FragmentIQ measures the error, by running the standard pipeline on synthetic muckpiles whose true fragmentation is known by construction. It performs the same watershed delineation and Rosin-Rammler fit as the classical tools, reports the recovered P_{10} , P_{50} and P_{80} , and compares them to the ground truth over seven cases spanning coarse to fine and monosized to broad. On that footing it quantifies the two biases and shows that one of them is largely correctable, and it tests whether a learned edge-delineation, in place of the classical watershed, helps. The contribution is not a new fragmentation algorithm; it is the honest, ground-truthed quantification of the bias of the standard one, and the demonstration that a size calibration removes most of it.

II. THE SYSTEMATIC BIAS

The core result is Fig. 1, and it is a clean confirmation of what the literature warns of, now with numbers. On every one of the seven synthetic muckpiles, the recovered P_{50} and P_{80} fall below the true values (Fig. 1a): the pipeline reports a muckpile finer than it really is, by roughly twenty to thirty percent, and the under-recovery is worst for the coarse cases (a true P_{80} of 1096 mm is recovered as 794). The mechanism is visible in Fig. 1b: the watershed *over-segments*, finding between one and a half and more than two times as

F. Santibañez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanez@gmail.com; ORCID 0000-0002-0150-3246).

Technical report, 2026. Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_FragmentIQ; live at <https://fragmentiq.fasl-work.com>.

TABLE I

RECOVERED VERSUS TRUE FRAGMENTATION OVER THE SEVEN SYNTHETIC MUCKPILES (T/R = TRUE/RECOVERED): THE WATERSHED FINDS MORE FRAGMENTS THAN EXIST AND RECOVERS P_{50} AND P_{80} BELOW TRUTH IN EVERY CASE.

Case	regime	found/true	P_{50} t/r (mm)	P_{80} t/r (mm)
R-COARSE	coarse	116/70	836/698	1096/794
R-MEDIUM	medium	145/98	356/260	416/310
R-FINE	fine	211/94	164/126	188/144
I-EVEN	medium	163/81	310/245	374/274
I-SHADOW	medium	157/81	310/237	374/271
C-MONO	mono	221/107	120/97	120/104
C-KNOWN	known	173/68	397/272	611/458

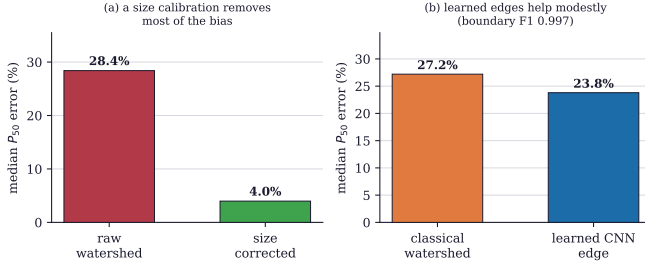


Fig. 2. Correcting the bias. (a) A size calibration that accounts for the over-segmentation and the missed fines reduces the median P_{50} error from 28.4% (raw watershed) to 3.98%, most of the bias is systematic and therefore correctable. (b) Replacing the classical watershed with a learned convolutional edge-delineation trims the error modestly, from 27.2% to 23.8%, at a boundary F1 of 0.997; better edges help, but the dominant fix is the size calibration, not the delineation.

many fragments as truly exist, because it splits single large blocks along internal texture and shadow into several apparent fragments. More, smaller apparent fragments drag the size distribution fine, which is exactly the direction of the observed bias. This is not a coding error, it is the intrinsic behaviour of watershed delineation on rock imagery, and quantifying it, rather than warning about it in the abstract, is the point: a raw image-based P_{50} should be read as an underestimate of tens of percent.

III. CORRECTING THE BIAS

The bias being systematic, it is largely correctable (Fig. 2). A size calibration, fit on a disjoint tuning set and accounting for both the over-segmentation and the fines that fall below the image resolution, reduces the median P_{50} error from 28.4% for the raw watershed to 3.98% (Fig. 2a), an order-of-magnitude reduction, because the bias is a consistent fine-shift that a single calibration can undo. The second lever, a better delineator, helps less. Replacing the classical watershed with a learned convolutional edge-detector (achieving a boundary F1 of 0.997 against the true fragment outlines) lowers the raw P_{50} error from 27.2% to 23.8% (Fig. 2b), a real but modest gain: the learned edges do split fewer blocks spuriously, but they do not, on their own, close the gap the way the calibration does. The honest ranking of fixes matters for practice: invest in calibrating the size estimate against a known reference, and treat a better segmentation as a secondary improvement, not the cure.

IV. DISCUSSION AND SCOPE

The report's value is a number where the field usually has a caveat. Image-based fragmentation analysis is biased fine, here by twenty to thirty percent at P_{50} and P_{80} , driven by watershed over-segmentation, and that bias is systematic enough that a size calibration removes almost all of it, taking the P_{50} error to about four percent, while a learned segmentation helps only modestly. A practitioner should take from this that a raw muckpile image number is an underestimate to be calibrated, not a measurement to be trusted, and that the calibration, not the segmentation method, is where accuracy is won. The scope is stated plainly. The seven cases are synthetic muckpiles, used precisely because their true size distribution is known and the bias can therefore be measured, which is not possible on a real muckpile without independent sieving; the magnitudes here are representative of the mechanism, not a calibration transferable to a specific site, which would need its own ground-truthed reference. The tool has no photo upload in this build; the cases are the inputs, and it is a didactic and diagnostic instrument, not a production granulometry system. None of this is hidden; the tool shows the recovered curve, the fit and the caveats on every case.

V. CONCLUSION

Measured against ground truth on synthetic muckpiles, image-based fragmentation analysis by watershed delineation is biased fine: it over-segments, finding up to twice the true fragment count, and recovers P_{50} and P_{80} some twenty to thirty percent below truth. The bias is systematic, so a size calibration reduces the median P_{50} error from 28.4% to 3.98%, while a learned edge-delineation helps only modestly (27.2% to 23.8%). Quantifying the bias where the literature warns of it, and showing that calibration, not segmentation, is the dominant fix, is the report's contribution, on a reproducible tool that makes the bias and its correction visible.

APPENDIX A REPRODUCTION

The committed artifacts under `data/derived/` carry the seven cases' true and recovered size distributions and fragment counts (`case-results.json`) and the learned-model metrics (`fq-learned.json`: the raw and size-corrected P_{50} errors, and the classical versus learned edge-delineation errors with the boundary F1). The figure scripts under `manuscripts/fragmentation/figures/` read those artifacts. The synthetic generator, watershed and fit are deterministic given a case seed, so re-running reproduces Figs. 1–2.

REFERENCES

- [1] V. M. Kuznetsov, "The mean diameter of the fragments formed by blasting rock," *Soviet Mining Science*, vol. 9, no. 2, pp. 144–148, 1973. doi:10.1007/BF02506177.
- [2] C. V. B. Cunningham, "Fragmentation estimations and the Kuz-Ram model, four years on," in *Proc. 2nd Int. Symp. on Rock Fragmentation by Blasting*, Keystone, CO, 1987, pp. 475–487.
- [3] P. Rosin and E. Rammler, "The laws governing the fineness of powdered coal," *Journal of the Institute of Fuel*, vol. 7, pp. 29–36, 1933.

- [4] L. Vincent and P. Soille, "Watersheds in digital spaces: An efficient algorithm based on immersion simulations," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 13, no. 6, pp. 583–598, 1991. doi:10.1109/34.87344.
- [5] N. H. Maerz, T. C. Palangio, and J. A. Franklin, "WipFrag image based granulometry system," in *Proc. FRAGBLAST 5 Workshop on Measurement of Blast Fragmentation*, Montreal, 1996, pp. 91–99.
- [6] W. Qiao *et al.*, "Deep learning-based pixel-level rock fragment recognition during tunnel excavation using instance segmentation model," *Tunnelling and Underground Space Technology*, vol. 115, art. 104072, 2021. doi:10.1016/j.tust.2021.104072.